# 1

## i

Our goal is to show that the edges in $T^*$ are all the same (although the edge weights may be different).

If $T$ is an optimal message-passing tree then $T$ is an MST of a graph whose edge weights are $w_e = p_e$

If $T$ is a MST of a graph whose edge weights are $w_e = p_e$ then $T$ is an optimal message-passing tree

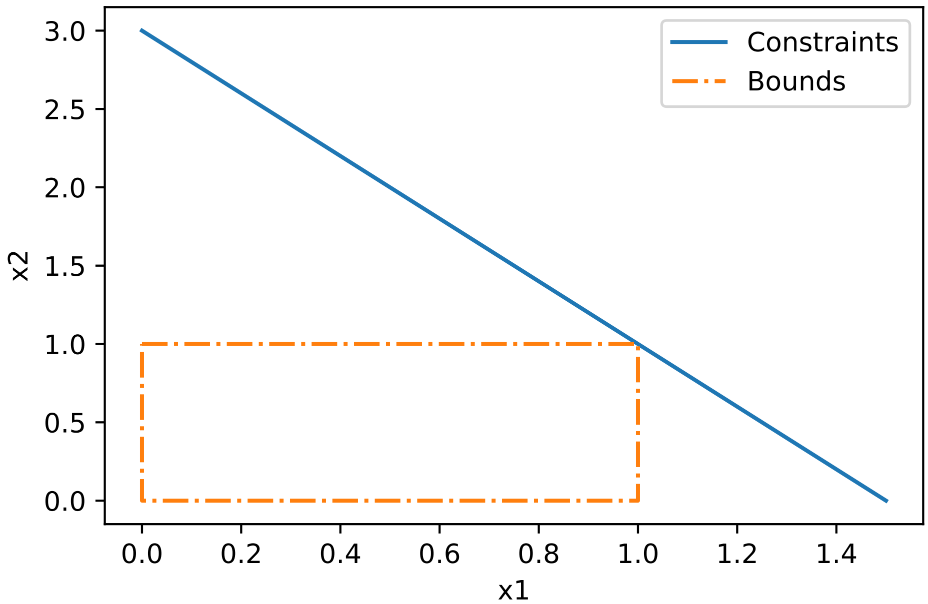
If $T$ is not an optimal message-passing tree then $T$ is not a MST of a graph whose edge weights are $w_e = p_e$

If $T$ is not a MST of a graph whose edge weights are $w_e = p_e$ then $T$ is not an optimal message-passing tree

## ii

# 2

## a



This program has a unique optimal solution which is $x^* = (x_1 = 1, x_2 = 1)$

### b

We are given: $\max x_1 + x_2$

s.t. $2x_1 + x_2 \leq 3$

$0 \leq x_1, x_2 \leq 1$

Thus the standard form is:

$\min -y_1 - y_2$

s.t. $2(y_1) + (y_2) + s_1 = 3$

$y_1 + z_1 = 1$

$y_2 + z_2 = 1$

$y_1, z_1, y_2, z_2, s_1 \geq 0$

### c

```
Maximize
  obj: x1 + x2
Subject To
  c1: 2x1 + x2 <= 3
Bounds
  0 <= x1 <= 1
  0 <= x2 <= 1
End
```

## 3

### a

### b

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$c = \begin{pmatrix} -1 \\ -(1-\lambda) \end{pmatrix}$$

$$A = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$b = (3)$$

## c

### d

# 4

# 5

### a

### b

### c

### d

### e

# 6

### a

### b

### c

### d

### e

### f